Stochastic programming
Multi-stage problems

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Motivation: example

- Assume that we invest in the market. We would like to reach a nominal goal G after Y years.
- The portfolio is rebalanced every v years, so there are $T = Y/v$ times where we have to decide what are the investments to do.
  There are T stages in our problem.
- We face a space of N investment decisions (i.e. we have N possible actions); we have a set $\mathcal{T} = \{1, 2, \dots, T\}$ of investment periods.
- If we go over the target G, we will have an interest rate q on the surplus.
- If we do not meet the target, we have to borrow the difference at an interest r.
- The initial wealth is b.

Variables

- x_{it}, $i \in N$, $t \in \mathcal{T}$: amount of money to invest in action i during the period t;
- y: excess amount at the end of the horizon;
- w: lack of money at the end of the horizon;
- In the current framework, we aim to take the corrective actions in order to reach the target of G dollars.

Deterministic formulation

$$
\begin{aligned}
\max \ & qy - rw \\
\text{s.t.} \ & \sum_{i \in N} x_{i1} = b, \\
& \sum_{i \in N} \omega_{it} x_{i,t-1} = \sum_{i \in N} x_{it}, \quad \forall t \in \mathcal{T} \backslash 1, \\
& \sum_{i \in N} \omega_{iT} x_{iT} - y + w = G, \\
& x_{it} \geq 0, \quad \forall i \in N, \ t \in \mathcal{T}, \\
& y, w \geq 0.
\end{aligned}
$$

Random returns

- The investment returns can be viewed as random variables, as we do not know their realizations in advance.
- Imagine that for each action, at each time t, there are R possible realizations.
- The scenarios are made of all the possible sequences of realizations.

Example: assume $R = 4$ and $T = 3$. Scenarios:

$t = 1$	$t = 2$	$t = 3$
1	1	1
1	1	2
1	1	3
1	1	4
1	2	1
$\vdots$	$\vdots$	$\vdots$
4	4	4

Stochastic program formulation

- Let x_{its}, $t \in \mathcal{T}$, $s \in S$ be the amount to invest in action i during the period t in scenario s;
- y_s: excess amount at the end of the horizon for scenario s;
- w_s: lack of money at the end of the horizon for scenario s.
- Let ω_{it} be the random return; it can be seen as function of the scenario s.

Stochastic version

$$
\begin{aligned}
\max \sum_{s=1}^{S} p_s(qy_s - rw_s) \\
\text{s.t.} \sum_{i \in N} x_{i1} &= b, \\
\sum_{i \in N} \omega_{its} x_{i,t-1,s} &= \sum_{i \in N} x_{its}, \quad \forall t \in \mathcal{T} \setminus 1, \forall s \in S \\
\sum_{i \in N} \omega_{iTs} x_{iT} - y_s + w_s &= G, \quad \forall s \in S, \\
x_{its} &\geq 0, \quad \forall i \in N,\ t \in \mathcal{T}, \forall s \in S, \\
y_s, w_s &\geq 0 \quad \forall s \in S.
\end{aligned}
$$

Scenario tree

Conceptually, the sequence of events can be arranged in a tree. There is one leaf per scenario.

The scenarios can have different probabilities, the variables can be correlated,. . .

Non-anticipativity

- The conversion between the deterministic model and the stochastic version is easy…
- but the built model is wrong!!!
- We have to enforce the non-anticipativity:

Let S_s^t be the set of scenarios that are identical to scenario s at time t. We have to guarantee that

$$x_{its} = x_{its'}, \ \forall i \in N, \ \forall t \in T, \ \forall s \in S, \ \forall s' \in S_s^t.$$

Non-anticipativity (cont'd)

Equivalently,

$$\left(\sum_{s' \in S_s^t} p'_s\right) x_{its} = \sum_{s' \in S_s^t} p'_s x_{its'} \quad \forall\, i, t, s,$$

or

$$x_{its} = \frac{\sum_{s' \in S_s^t} p'_s x_{its'}}{\sum_{s' \in S_s^t} p'_s} \quad \forall\, i, t, s$$
$$= E[x_{its'} | s' \in S_s^t] \quad \forall\, i, t, s.$$

The amount invested in scenario s, multiplied by the probability to reach a scenario equivalent to s, has to be equal to the expected amount that would be invested in any scenario equivalent to s.
There are many formulations of these conditions!

Back to the scenario tree

- We have to enforce nonanticipativity as we have created copies of the variables for each scenario, at each period. Graphically:

Nodal approach

- We can also create a vector of variables for each node in the tree.
- This vector corresponds to what should be our decision given the known realizations of the random variables upon this stage (i.e. at this node).
- Index the nodes by $l = 1, 2, \ldots, \mathcal{L}$.
- We have to know the ancestor of each node. Let $A(l)$ be the ancestor of node $l \in \mathcal{L}$ in the scenarios tree.

Reformulation

$$
\begin{aligned}
\max \; & \sum_{s\in S} p_s(q y_s && - r w_s) \\
\text{s.t.} \; & \sum_{i\in N} x_{i1} && = b, \\
& \sum_{i\in N} \omega_{il} x_{i,A(l)} && = \sum_{i\in N} x_{il}, \quad \forall l \in \mathcal{L}\backslash 1, \\
& \sum_{i\in N} w_{iA(s)} x_{iA(s)} - y_s + w_s && = G, \quad \forall s \in S, \\
& x_{il} && \geq 0, \quad \forall i \in N, \; \forall l \in \mathcal{L}, \\
& y_s, w_s && \geq 0 \quad \forall s \in S.
\end{aligned}
$$

Multi-period production planning

- A firm produces various different goods.
- Resources (for instances manual and machine work hours) are required. We assume that these requirements are known for each product.
- The demand must be met at the end of each period, but this demand is not perfectly known.
- A too large or too small inventory induces costs (storage, additional purchases to satisfy the demand,. . .).
- It is possible to add machine and work hours, within predefined limits.
- There is a hiring and firing cost related to changes in the workforce.

Decision problem

- We have to decide now on
  - the quantity of each product to manufacture during each period;
  - the additional capacity to use during each period;
  - to hire or to fire staff during each period;
- A random demand occurs. Conceptually, this happens at each period.
- After having observed the random demands, we can decide on how to stock the products in the inventory, or to buy from an external source.

Definitions

1. Sets:
    - T: number of periods, also seen as a set;
    - N: products;
    - M: resources.
2. Variables:
    - x_{jt}: production of product $j \in N$ during the period $t \in T$;
    - u_{it}: additional amount of resource $i \in M$ to buy during the period $t \in T$;
    - $z^{+}_{t-1,t}$, $z^{-}_{t-1,t}$: planned augmentation or reduction of the labor force;
    - $y^{+}_{j,t}$, $y^{-}_{j,t}$: surplus, deficit of product $j \in N$ at the end of the period $t \in T$.

Parameters

- All the above variables have associated costs $(\alpha, \beta, \gamma, \delta)$.
- ω_{jt}: demand of product $j \in N$ during the period $t \in T$.
- U_{it}: upper bound on u_{it}.
- a_{ij}: amount of resource $i \in M$ required to produce one unit of the good $j \in N$.
- b_{it}: amount of resource $i \in M$ available at time $t \in T$.

Model

$$\min \sum_{j\in N}\sum_{t\in T}\alpha_{jt}x_{jt} + \sum_{i\in M}\sum_{t\in T}\beta_{it}u_{it} + \sum_{t\in T\setminus 1}\left(\gamma^{+}_{t-1,t}z^{+}_{t-1,t} + \gamma^{-}_{t-1,t}z^{-}_{t-1,t}\right)$$

$$+ E_{\omega}\left[\min\left(\sum_{j\in N}\sum_{t\in T}(\delta^{+}_{jt}y^{+}_{jt} + \delta^{-}_{jt}y^{-}_{jt})\right)\right],$$

such that

$$\sum_{j\in N} a_{ij}x_{jt} \leq b_{it} + u_{it} \quad \forall i\in M,\ \forall t\in T,$$

$$u_{it} \leq U_{it} \qquad \forall i\in M,\ \forall t\in T,$$

$$z^{+}_{t-1,t} - z^{-}_{t-1,t} = \sum_{j\in N} a_{jt}(x_{jt} - x_{j,t-1}) \quad \forall t\in T\setminus 1,$$

$$x_{jt} + y^{+}_{j,t-1} - y^{+}_{jt} + y^{-}_{jt} = \omega_{jt} \qquad \forall j\in N,\ \forall t\in T \text{ (demand satisfaction)},$$

Extended form of the recourse model

$$\min \sum_{j\in N}\sum_{t\in T}\alpha_{jt}\boldsymbol{x}_{jt} + \sum_{i\in M}\sum_{t\in T}\beta_{it}\boldsymbol{u}_{it} + \sum_{t\in T\backslash 1}\left(\gamma^+_{t-1,t}\boldsymbol{z}^+_{t-1,t} + \gamma^-_{t-1,t}\boldsymbol{z}^-_{t-1,t}\right)$$

$$+ \color{red}{\sum_{s\in S}}\color{black}\sum_{j\in N}\sum_{t\in T} p_s(\delta^+_{jt}y^+_{jt\color{red}{s}} + \delta^-_{jt}y^-_{jt\color{red}{s}}),$$

such that

$$\sum_{j\in N} a_{ij}\boldsymbol{x}_{jt} \le b_{it} + \boldsymbol{u}_{it} \quad \forall i\in M,\ \forall t\in T,$$

$$\boldsymbol{u}_{it} \le U_{it} \qquad \forall i\in M,\ \forall t\in T,$$

$$\boldsymbol{z}^+_{t-1,t} - \boldsymbol{z}^-_{t-1,t} = \sum_{j\in N} a_{jt}(\boldsymbol{x}_{jt} - \boldsymbol{x}_{j,t-1}) \quad \forall t\in T\backslash 1,$$

$$\boldsymbol{x}_{jt} + y^+_{j,t-1,\color{red}{s}} - y^+_{jt\color{red}{s}} + y^-_{jt\color{red}{s}} = \omega_{jt\color{red}{s}} \qquad \forall j\in N,\ \forall t\in T,$$

nonanticipativity constraints.

Potential solution techniques

Adapted from Jalal Kazempour,
`https://www.jalalkazempour.com/teaching`

- Direct solution: computational issues
- Linear decision rules: a non-decomposition approach with approximation. Assumption of linear dependency between decision variables and uncertain parameters. Using this approximation, the number of constraints and variables grows only linearly with the number of stages.
- Nested Benders decomposition: a decomposition approach based on fixing complicating variables.
- Lagrangian relaxation: a decomposition approach based on relaxing the complicating (non-anticipativity) constraints.

Nested Benders decomposition

- Nested Benders decomposition, also called the multistage L-shaped method, generalizes the classical two-stage Benders decomposition to stochastic linear programs with multiple decision stages.
- It recursively applies (two-stage) Benders decomposition on a scenario tree, where each stage generates cuts for its parent stage.

Model structure

Consider a T-stage stochastic linear program of the form:

$$
\begin{aligned}
\min_{x_1,\dots,x_T} \quad & \mathbb{E}\left[\sum_{t=1}^{T} c_t(\xi_t)^\top x_t\right] \\
\text{s.t.} \quad & A_1 x_1 = b_1, \quad x_1 \geq 0, \\
& A_t x_t = b_t(\xi_t) - T_t(\xi_t) x_{t-1}, \quad x_t \geq 0, \quad t = 2, \dots, T.
\end{aligned}
$$

Each ξ_t represents the random information revealed at stage t, defining a scenario tree.

Recursive value functions

Define recursively the cost-to-go function:

$$Q_t(x_{t-1}, \xi_t) = \min_{x_t \geq 0} c_t(\xi_t)^\top x_t + \mathbb{E}[Q_{t+1}(x_t, \xi_{t+1}) \mid \xi_t]$$
$$\text{s.t. } A_t x_t = b_t(\xi_t) - T_t(\xi_t) x_{t-1},$$

with $Q_{T+1} \equiv 0$. The goal is to approximate Q_t at each stage with a piecewise-linear function constructed from Benders cuts.

Assume for simplicity the recourse is complete.

Cut generation (optimality cut)

At iteration k, for each scenario ξ_t^s, the subproblem is solved:

$$\begin{aligned} \min_{x_t^s \geq 0} \quad & c_t^{s\top} x_t^s + \sum_{s'} p_{s'|s} \theta_{t+1}^{s'} \\ \text{s.t.} \quad & A_t^s x_t^s = b_t^s - T_t^s x_{t-1}^{p(s)}. \end{aligned}$$

Let π_t^s denote the optimal duals for the equality constraints. Then the expected cost-to-go can be approximated as:

$$Q_t(x_{t-1}) \geq \alpha_t + \beta_t^\top x_{t-1}, \tag{1}$$

where

$$\alpha_t = \sum_s p_s (\pi_t^s)^\top b_t^s, \qquad \beta_t = -\sum_s p_s (\pi_t^s)^\top T_t^s. \tag{2}$$

These (α_t, β_t) pairs define the Benders cuts for the parent stage.

Algorithm outline

1. **Initialization:** Start with initial lower approximations θ_t for all stages.
2. **Forward pass:** Fix first-stage decision x_1 and recursively compute trial points x_t through the scenario tree.
3. **Backward pass:** Starting from the leaves, solve subproblems, extract duals, and form cuts for each parent node.
4. **Master problem update:** Add the new cuts to each parent's master problem and reoptimize.
5. **Convergence:** Stop when the expected gap between upper and lower bounds is within tolerance.

Remarks

- The algorithm is naturally parallelizable across scenarios.
- Feasibility cuts are added if subproblems are infeasible, similar to two-stage Benders.
- For convex piecewise-linear approximations, monotonic convergence to the optimal policy is guaranteed.

Illustration

Step 1: forward pass

Step 1: forward pass

Solve

$$\begin{aligned}
\min_{x_1,\theta_1} \;& c_1^T x_1 + \theta_1 \\
\text{s.t. } & A_1 x_1 = b_1 \\
& \theta_1 \geq \theta_{1,\min} \\
& \text{feasibility and optimality cuts} \\
& x_1 \geq 0.
\end{aligned}$$

Step 2: forward pass

Step 2: forward pass

Solve

$$\begin{aligned}\min_{x_{2,i},\theta_{2,i}} \quad & c_{2,i}^T x_{2,i} + \theta_{2,i} \\ \text{s.t.} \quad & A_{2,i} x_{2,i} = h_{2,i} - T_{2,i} x_0 \\ & \theta_{2,i} \geq \theta_{2,i,\min} \\ & \text{feasibility and optimality cuts} \\ & x_{2,i} \geq 0.\end{aligned}$$

Step 3: forward pass

Step 2: backward pass

The newly formed cuts are added to the Step 2 program and we solve again the program to determine cuts on the second-stage program.

Step 2: backward pass

Step 1: backward pass

The newly formed cuts are added to the Step 1 program.

This forms a new master program for the first stage, that is solved again in the forward pass.

Step 1: backward pass

Scenario tree

The main issue is the construction of the scenario tree that is growing exponentially at each stage, and so the number of subproblems.

The stochastic dual dynamic programming algorithm aims to solve the problem on a lattice rather than a tree, under the assumption that the underlying random process is Markovian.

The augmented Lagrangian approaches will still rely on the scenario tree, but will consider the relaxation on the nonanticipativity constraints rather than solve a succession a two-stage problems. This limits the number of subproblems to the number of scenarios and can handle non-Markovian processes. The drawback is that it can take many iterations to restore non-anticipativity.